

Regime-Dependent Contagion in DeFi Liquidations: Disentangling Self-Excitation from Common Price Shocks

Vihan Shah

Adani International School, Ahmedabad

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Abstract

Liquidation events in decentralized lending protocols cluster sharply in time, but clustering alone cannot distinguish endogenous contagion from common exposure to shared price shocks. Using 18,408 ETH-collateralized liquidations on Aave V3 (January 2023–June 2026), I model hourly event counts as a negative-binomial self-exciting count process with a price-driven background intensity and a regime-dependent excitation term. Three findings emerge. First, a naive Hawkes model with no price control is estimated to be supercritical ($\nu = 1.18$), a known artifact of unmodeled background nonstationarity. Second, once the common price shock is controlled for, normal-regime contagion is subcritical: because the contemporaneous control over-corrects (absorbing genuine feedback) while a lagged control under-corrects (missing same-hour shocks), the two specifications bracket the normal-regime branching ratio in $[0.69, 0.79]$. Third, contagion is significantly regime-dependent (likelihood-ratio $p < 10^{-5}$): during stress episodes, defined ex ante by trailing 24-hour ETH drawdowns exceeding 7%, the bracketed point estimates rise to $[1.12, 1.81]$, at or above the supercritical boundary. Unconditional risk measures therefore misstate average contagion and, more importantly, understate systemic instability precisely in the episodes where cascades are most likely.

1 Introduction

Decentralized lending protocols such as Aave allow users to borrow assets against posted collateral. When the value of a borrower’s collateral falls below a threshold, the position is liquidated: a third party repays the debt and seizes the collateral at a discount. Because liquidations sell collateral into the market, they can depress the very price that determines the solvency of other positions. This creates the potential for contagion—a liquidation mechanically triggering further liquidations—a mechanism widely cited as a source of systemic risk in decentralized finance (DeFi) [4, 5].

Empirically, liquidations cluster in time. Identifying the data-generating process matters because clustering is consistent with two very different mechanisms: genuine self-exciting contagion, or a common response to an exogenous price drop. The two have opposite policy implications—the first calls for protocol-level circuit breakers on the liquidation mechanism itself, the second merely reflects correlated exposure.

This paper separates the two channels using a self-exciting count process with an explicit control for the common price shock, and asks whether the endogenous component varies with market conditions. The identification problem is subtle in both directions. Omitting the price control loads common-shock co-movement onto the excitation term and biases contagion estimates *upward*—in this sample, all the way past the critical boundary. But a contemporaneous price control introduces the opposite bias: forced collateral sales depress the same-hour price, so the control is partly an *outcome* of the liquidations it is meant to purge, and the estimator over-attributes genuine feedback to the price term, biasing contagion *downward*. A lagged control avoids the simultaneity but leaves the same-hour shock unabsorbed, restoring some upward bias. Rather than privileging either specification, I report both and interpret them as a bracket on the true branching ratio.

The central result is regime dependence. In normal conditions the bracketed branching ratio is $[0.69, 0.79]$ —a substantial but subcritical endogenous component. In stress episodes, identified ex ante from price drawdowns, the bracket rises to $[1.12, 1.81]$, at or above the supercritical boundary at which each liquidation triggers, on average, at least one more. A likelihood-ratio test rejects regime homogeneity at $p < 10^{-5}$.

2 Data and Sample Construction

I analyze 18,408 liquidations on Aave V3 (Ethereum), extracted via Dune Analytics (query ID: 7814234). The sample covers January 30, 2023 to June 25, 2026, a span of 1,243 days and 29,833 observation hours. To ensure the common-shock control is exactly matched to the sample dynamics, collateral is restricted to ETH-denominated assets (WETH, wstETH, and cbETH). Contemporaneous ETH prices are hourly closes from Coinbase; 29 hours affected by API maintenance windows are forward-filled.

The series is extremely bursty: 91.4% of observation hours contain no liquidations, while the busiest single hour contains 532. The variance-to-mean ratio is 135.8, indicating severe overdispersion, and the busiest 1% of hours account for 78.8% of all events (Figures 1 and 2).

Technical hurdle: timestamp ties. The raw data contain many events with identical block timestamps. A continuous-time estimator interprets these ties as near-instantaneous self-excitation and returns a spurious sub-second decay. Aggregating events into hourly bins resolves this artifact, ensuring the estimator reflects protocol dynamics rather than block-level ties.

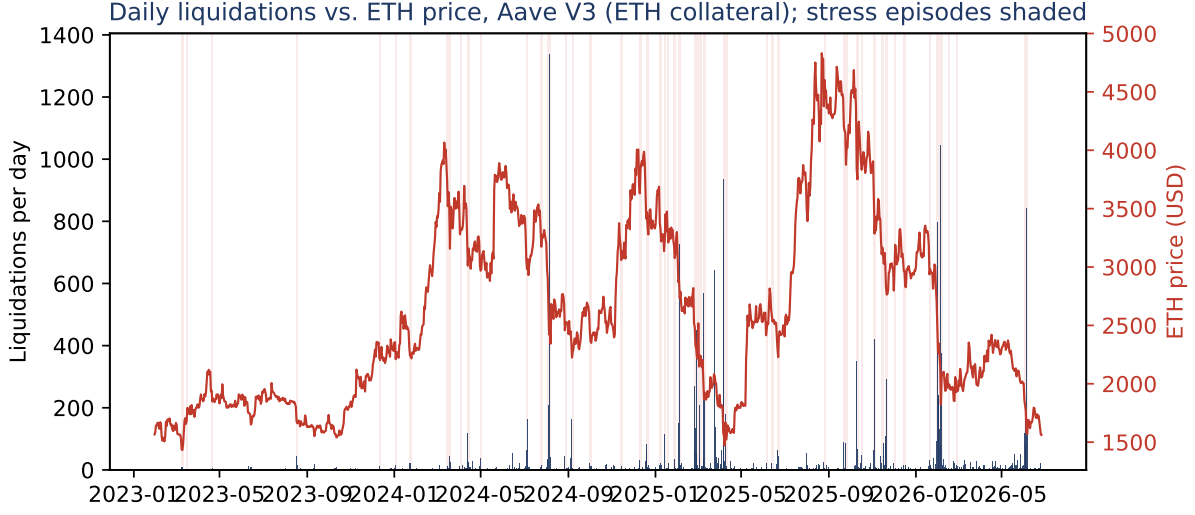


Figure 1: Daily liquidations (left axis, bars) vs. ETH price (right axis, line), Aave V3 ETH-collateral. Shaded bands mark stress episodes, defined ex ante as hours within ± 12 hours of a trailing 24-hour ETH return below -7% . Liquidation spikes align with sharp drawdowns, illustrating the common-shock channel that any contagion estimate must control for.

3 Model Specification

Let y_k denote the liquidation count in hour $k \in \{1, \dots, T\}$. The conditional mean intensity is

$$\lambda_k = \mu + \gamma s_k + \alpha(k) \sum_{j=1}^{k-1} e^{-\beta(k-j)} y_j, \quad (1)$$

where $s_k = \max(0, -r_k)$ is the downside hourly log return of ETH, γ scales the common-shock effect, β governs the decay of excitation, and $\alpha(k)$ is the self-excitation loading. In the pooled specification $\alpha(k) \equiv \alpha$; in the regime-dependent specification

$$\alpha(k) = \alpha_N \mathbf{1}\{k \in \text{normal}\} + \alpha_S \mathbf{1}\{k \in \text{stress}\}. \quad (2)$$

The branching ratio in regime $g \in \{N, S\}$ is $\nu_g = \alpha_g / (e^\beta - 1)$, the expected number of liquidations directly triggered by a single event. Because the counts are severely overdispersed, the model is estimated by maximum likelihood under a negative-binomial (NB2) distribution with dispersion parameter r , so that $\text{Var}(y_k | \mathcal{F}_{k-1}) = \lambda_k + \lambda_k^2 / r$. Confidence intervals are percentile intervals from a moving-block bootstrap with weekly (168-hour) blocks, preserving temporal dependence (250 replications; 200 for the naive specification).

Regime definition. A stress episode comprises all hours within ± 12 hours of any hour in which the trailing 24-hour ETH return falls below -7% . This yields 2,281 stress hours (7.6% of the sample). Two features of the rule matter for identification. First, it is defined on *price*,

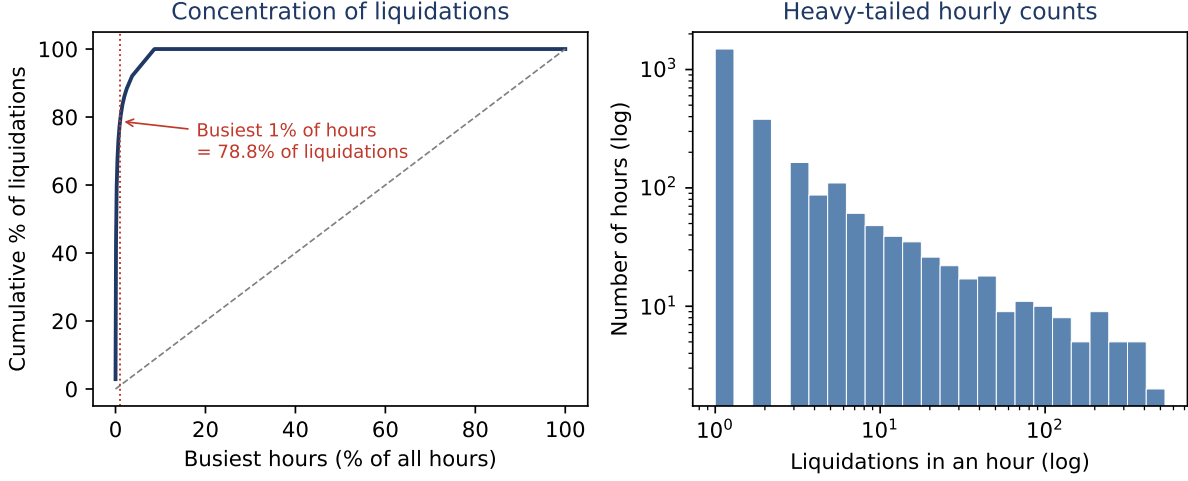


Figure 2: Clustering diagnostics. Left: the busiest 1% of hours account for 78.8% of all events. Right: distribution of hourly counts on log–log axes, confirming heavy-tailed dynamics.

not on liquidation activity, so the regime indicator does not condition on the outcome variable. Second, it is computable in real time from the trailing price path, so the classification is *ex ante* rather than selected with hindsight.

Endogeneity and the bracketing strategy. The contemporaneous control s_k is subject to simultaneity: forced collateral sales depress the same-hour price, so part of s_k is caused by the liquidations it controls for, and the price term absorbs genuine feedback—biasing $\hat{\nu}$ *downward*. Replacing s_k with the lagged return s_{k-1} removes the simultaneity but leaves the same-hour common shock uncontrolled, part of which then loads onto the excitation term—biasing $\hat{\nu}$ *upward*. Neither specification is unbiased, but their biases have opposite signs, so the pair brackets the true branching ratio. All regime results below are therefore reported under both controls.

4 Results

4.1 Goodness of Fit

A Poisson specification fits poorly, with a Pearson-residual standard deviation of 7.6 against a target of 1. The negative-binomial specification ($\hat{r} \approx 0.11$) centers the residuals (mean -0.001); excluding the ten most extreme hours, the residual standard deviation is 0.94. The NB2 model thus captures the heavy-tailed conditional distribution well, with only the most extreme cascade hours remaining in the tail.

Table 1: Maximum-likelihood parameter estimates (hourly bins, NB2 likelihood).

	Naive (no control)	Price-controlled (pooled)	Regime, contemp. control s_k	Regime, lagged control s_{k-1}
μ (baseline, events/h)	0.043	0.018	0.020	0.024
γ (price loading)	—	24.1	23.3	16.1
α_N (excitation, normal)	0.170	0.182	0.156	0.109
α_S (excitation, stress)	—	—	0.253	0.250
β (decay, h^{-1})	0.135	0.206	0.204	0.130
r (NB dispersion)	0.082	0.111	0.113	0.091
Excitation half-life (h)	5.1	3.4	3.4	5.3
Negative log-likelihood	12,509.2	12,008.7	11,997.3	12,318.3

Notes: $T = 29,833$ hours. In the naive and pooled columns α_N is the single excitation loading α . Estimation by Nelder–Mead on log-transformed parameters with multiple restarts. Bootstrap uncertainty for the branching ratios, the objects of interest, is reported in Table 2.

Table 2: Branching-ratio estimates (hourly bins; 95% moving-block-bootstrap percentile CIs).

Specification	Regime	$\hat{\nu}$	95% CI
Naive (no price control)	pooled	1.18	[0.94, 1.56]
Price-controlled, contemporaneous	pooled	0.80	[0.55, 1.32]
Price-controlled, contemporaneous	normal	0.69	[0.31, 1.32]
Price-controlled, lagged	normal	0.79	[0.45, 1.20]
Price-controlled, contemporaneous	stress	1.12	[0.89, 1.77]
Price-controlled, lagged	stress	1.81	[1.54, 2.88]

4.2 Main Estimates: The Price Control and the Bracket

Table 1 reports full parameter estimates; Table 2 and Figure 3 summarize the branching ratios.

The naive model with no price control is estimated to be supercritical: $\hat{\nu} = 1.18$ [0.94, 1.56]. Taken literally this implies an explosive process, which the data plainly are not; the estimate instead reflects a well-documented artifact. [2] show that when the background intensity is nonstationary and unmodeled, Hawkes estimators absorb the background variation into the excitation kernel and are driven toward (and past) the critical boundary. The price-driven baseline in Equation (1) models exactly the dominant source of nonstationarity here, and adding it moves the full-sample estimate to $\hat{\nu} = 0.80$ [0.55, 1.32] and roughly halves the estimated common-shock loading’s burden on the excitation term. The naive estimate should therefore be read as an upper envelope produced by misspecification, not as a measurement of contagion.

In the regime-dependent specification, the normal-regime branching ratio is $\hat{\nu}_N = 0.69$ [0.31, 1.32] under the contemporaneous control and $\hat{\nu}_N = 0.79$ [0.45, 1.20] under the lagged control. Consistent with the bias analysis above, the lagged estimate exceeds the contemporaneous one; the two bracket the normal-regime branching ratio in [0.69, 0.79]. This is a large endogenous component—roughly seven in ten liquidations in normal times are triggered offspring rather than fresh exogenous arrivals—but it is subcritical: cascades damp out.

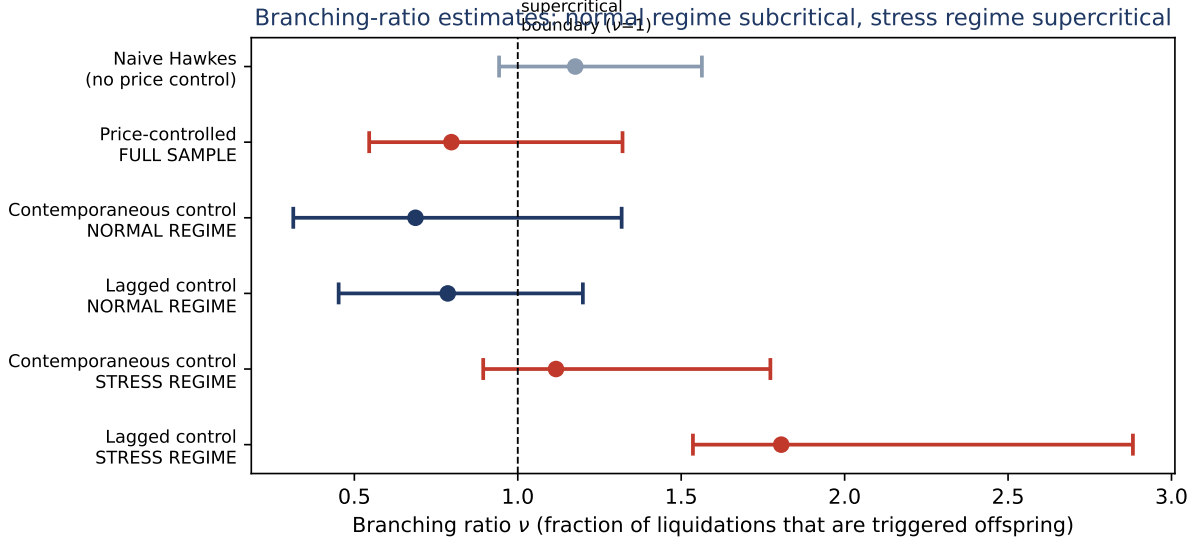


Figure 3: Branching-ratio estimates across specifications with 95% bootstrap CIs. The naive estimator is pushed past the supercritical boundary by unmodeled background nonstationarity. With the price control, the normal regime is subcritical (bracket $[0.69, 0.79]$) while stress-regime point estimates lie at or above criticality under both controls.

4.3 Regime Dependence

Allowing the excitation loading to differ across regimes improves the log-likelihood by 11.4 points for one additional parameter ($LR = 22.8$, $p = 1.8 \times 10^{-6}$): contagion intensity is significantly regime-dependent.

The stress-regime estimates are $\hat{\nu}_S = 1.12$ $[0.89, 1.77]$ under the contemporaneous control and $\hat{\nu}_S = 1.81$ $[1.54, 2.88]$ under the lagged control. The bracketing logic is asymmetric here in a way that strengthens the conclusion: the contemporaneous control is most contaminated—and therefore over-corrects most aggressively—precisely during crash hours, when liquidation-driven price impact is largest. That this *conservative* specification still returns a stress-regime point estimate above one is the key finding. The lagged specification, which rejects subcriticality outright, should be read as the upper edge of the bracket rather than a point measurement.

A locally supercritical branching ratio does not imply a literally explosive process: stress episodes are finite (7.6% of hours), and the process re-enters the subcritical normal regime as prices stabilize. The correct reading is that *within* stress windows the system operates at or above criticality, so cascade sizes are governed by near-critical branching dynamics—heavy-tailed and unpredictable—rather than by the damped geometric decay that average estimates imply.

4.4 Robustness to Bin Width

Figure 4 re-estimates the pooled specifications on 2-hour and 4-hour bins (sub-hourly bins are not testable here because the price control is sampled hourly). Two facts are stable across

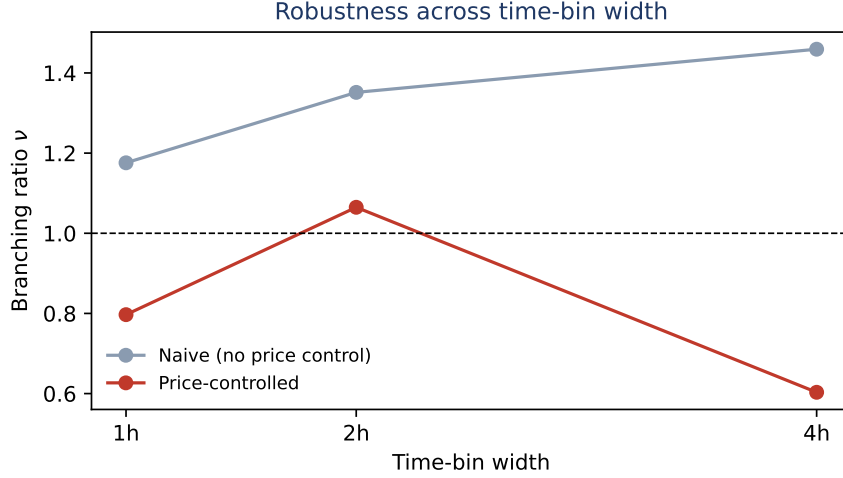


Figure 4: Robustness of the pooled branching ratio to time-bin width. The naive estimate is supercritical at every width; the price-controlled estimate lies below it at every width. Hourly bins are the primary specification, matching the sampling frequency of the price control.

widths. First, the naive estimate is supercritical at every bin width (1.18, 1.35, 1.46 at 1h, 2h, 4h), and drifts further from one as bins coarsen—consistent with coarser bins conflating more of the slow common price move with the excitation term. Second, the price-controlled estimate lies below the naive estimate at every width. The price-controlled point estimate itself is bin-sensitive (0.80, 1.07, 0.60), a known feature of discretized self-exciting models in which the bin width interacts with the kernel decay; the hourly specification is primary because it is the finest resolution at which the control variable is actually measured.

5 Discussion and Limitations

The results discipline two opposite errors in the systemic-risk literature on DeFi liquidations. Models that ignore common price factors overstate contagion—here, to the point of spurious supercriticality. But contagion is not merely a price artifact: after removing the common shock, an endogenous component of 0.69–0.79 survives in normal times, and it rises to or above the critical boundary in stress. Unconditional estimates average over the two regimes and are therefore miscalibrated for exactly the question that matters: how the system behaves when it is most vulnerable.

The proximity of several estimates to the critical boundary warrants particular caution in light of [2], who show that apparent criticality in Hawkes models frequently reflects misspecification—nonstationary baselines, edge effects, or kernel misfit—rather than true near-critical dynamics. Three features of the design partially mitigate this critique. The dominant source of baseline nonstationarity, the ETH price path, is modeled explicitly rather than left in the residual; the naive-versus-controlled comparison quantifies exactly how much apparent criticality that nonsta-

tionarity generates (it moves $\hat{\nu}$ from 1.18 to 0.80); and the regime split localizes the near-critical behavior to ex-ante-identified stress windows rather than attributing it to the sample as a whole. What the design cannot rule out is residual baseline variation *within* stress episodes—for example, oracle-update dynamics or gas-price spikes—that the single price covariate does not capture; the stress-regime estimates should be read with that reservation.

Further limitations qualify the conclusions. The price control is a reduced-form proxy for the common shock, not a structural model of it. The lagged specification removes simultaneity only under the assumption that liquidation price impact does not persist into the following hour; slower impact decay would push some feedback into the lagged control and re-introduce attenuation. The regime rule, while ex ante, involves two tuning constants (the -7% threshold and the ± 12 -hour window). The sample is restricted to ETH-denominated collateral on a single protocol, and cross-collateral or cross-protocol contagion—plausibly most active in stress—is outside the model. Finally, the supercritical stress-regime estimates are statements about local branching intensity, not predictions of unbounded cascades.

6 Conclusion

Liquidation contagion in Aave V3 is genuine, quantitatively large, and regime-dependent. A naive self-exciting model overstates it badly, to the point of spurious supercriticality; a contemporaneous price control understates it by absorbing genuine feedback. Bracketing between the two, roughly seven in ten liquidations in normal conditions are endogenously triggered—a subcritical but substantial multiplier. In stress episodes, identified ex ante from price draw-downs, the branching ratio rises to or above the critical boundary, and regime homogeneity is rejected at $p < 10^{-5}$. Risk measures built on unconditional clustering statistics are therefore miscalibrated twice over: they overstate everyday contagion while understating the near-critical cascade dynamics that emerge precisely when the system is most fragile.

Disclosure and Acknowledgement

In accordance with institutional policy: I used Claude (Anthropic) for code implementation, and manuscript formatting and Gemini 3 Pro for technical proofreading. All estimates reported here were generated directly from the posted data by the accompanying code, and I have verified and am fully accountable for the accuracy of the analysis. I thank Dune Analytics and Coinbase for data access.

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A Computational Appendix

Maximum-likelihood estimation was implemented in Python (SciPy Nelder–Mead) on log-transformed parameters with multiple random restarts. The recursive excitation term $E_k = \sum_{j < k} e^{-\beta(k-j)} y_j$ is computed in $O(T)$ via a first-order IIR filter:

```
def excitation(y, beta):
    d = np.exp(-beta)
    return lfilter([0.0, d], [1.0, -d], y)
```

Bootstrap confidence intervals resample the joint series $(y_k, s_k, \text{regime}_k)$ in moving blocks of 168 hours with replacement to the original sample length, re-estimating from the full-sample MLE as the starting point (250 replications; 200 for the naive specification); reported intervals are the 2.5th and 97.5th percentiles of the replicate branching ratios. Stress episodes are all hours within ± 12 hours of any hour whose trailing 24-hour ETH return is below -7% (2,281 hours; 7.6% of the sample). Dune Analytics query ID: 7814234.